PHY211 ASSIGNMENT

NOTE: Always define each parameter in each of your equations, and tag your relevant equations clearly

From the expression for the total energy of the electron as the sum of is kinetic and potential energy, *assuming that the electron is not moving at relativistic speeds*;

i. Clearly show that the orbital energies of a hydrogen-like atom or ion with atomic number Z is given by

$$E_n = -\frac{Z^2}{n^2} E_0$$

Where E_0 is the ground state energy.

ii. Show that the ground state energy for hydrogen atom is

E_0 = 13.6 eV

iii. Show that the energy of a photon of frequency υ emitted in a downward transition is given by

$$h\upsilon = (13.6eV)\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

Where subscripts *i* and *f* denote *initial* and *final* energy levels.

NOTE: *To be submitted by the next class*